

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

Product Name Cesium

CAS No 7440-46-2

Synonyms None

Product Code **10146**

Address ThermoFisher Scientific Australia Pty Ltd  
5 Caribbean Drive, Scoresby  
VICTORIA 3179, Australia

Emergency Tel. **CHEMTREC®**  
**03 9757 4559 or +613 9757 4559**

Telephone / Fax Numbers Tel: 1300 735 292  
Fax: 1800 067 639

E-mail address ANZinfo@thermofisher.com

Recommended Use Laboratory chemicals.

Uses advised against This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.  
This product does not contain any substance(s) subject to Prohibition, Authorization or  
Restriction. This product does not contain any substance(s) listed on the voluntary National  
Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

#### Physical hazards

Substances/mixtures which, in contact with water, emit flammable gases Category 1

#### Health hazards

Skin Corrosion/Irritation Category 1 B  
Serious Eye Damage/Eye Irritation Category 1  
Reproductive Toxicity Category 2

#### Environmental hazards

No hazards identified

#### Label Elements



Flame



Health Hazard



Corrosion

**Signal Word****Danger****Hazard Statements**

H260 - In contact with water releases flammable gases which may ignite spontaneously

H314 - Causes severe skin burns and eye damage

H361 - Suspected of damaging fertility or the unborn child

AUH014 - Reacts violently with water

**Precautionary Statements**

P201 - Obtain special instructions before use

P202 - Do not handle until all safety precautions have been read and understood

P223 - Do not allow contact with water

P231 + P232 - Handle and store contents under inert gas. Protect from moisture

P260 - Do not breathe dust/fume/gas/mist/vapors/spray

P264 - Wash face, hands and any exposed skin thoroughly after handling

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P363 - Wash contaminated clothing before reuse

P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

P302 + P335 + P334 - IF ON SKIN: Brush off loose particles from skin. Immerse in cool water

P402 + P404 - Store in a dry place. Store in a closed container

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information**

This product does not contain any known or suspected endocrine disruptors

## Section 3 - Composition and Information on Ingredients

| Component | CAS No    | Weight % |
|-----------|-----------|----------|
| Cesium    | 7440-46-2 | >95      |

## Section 4 - First Aid Measures

**Inhalation**

Remove from exposure, lie down. Remove to fresh air. If breathing is difficult, give oxygen. If not breathing, give artificial respiration. Immediate medical attention is required.

**Ingestion**

Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Drink plenty of water. Call a physician immediately. If possible drink milk afterwards.

**Skin Contact**

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Immediate medical attention is required.

|                                            |                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                         | Immediate medical attention is required. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.                                                                                                                                                               |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                            |
| <b>Most important symptoms and effects</b> | Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                                                                                                                                                                          |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

Dry chemical. approved class D extinguishers. clay. sodium carbonate.

### Extinguishing media which must not be used for safety reasons

Water. Carbon dioxide (CO<sub>2</sub>). Foam. Do not use halon type extinguisher.

### Hazardous Decomposition Products

Thermal decomposition can lead to release of irritating gases and vapors, Hydrogen.

### Specific Hazards Arising from the Chemical

Water reactive. Produce flammable gases on contact with water.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

### Emergency procedures

Ensure adequate ventilation.

### Environmental Precautions

See Section 12 for additional Ecological Information.

### Methods for Containment and Clean Up

#### Clean-up methods - small spillage

Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment. Sweep up and shovel into suitable containers for disposal.

#### Clean-up methods - large spillage

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling**

Do not breathe dust. Do not get in eyes, on skin, or on clothing. Handle product only in closed system or provide appropriate exhaust ventilation. Wash thoroughly after handling. Use spark-proof tools and explosion-proof equipment. Use only non-sparking tools. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Container should be opened by a technically qualified person.

**Conditions for Safe Storage, Including any Incompatibilities**

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep from any possible contact with water. Corrosives area. Store under an inert atmosphere. Keep away from water or moist air.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

**Exposure limits**

The product does not contain any hazardous materials with occupational exposure limits established.

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Exposure Controls****Engineering Measures**

Use explosion-proof electrical/ventilating/lighting equipment. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Natural rubber | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |
| Nitrile rubber |                                   |                 |                 |                       |
| Neoprene       |                                   |                 |                 |                       |
| PVC            |                                   |                 |                 |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection**

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:**

Particulates filter conforming to EN 143 (or AUS/NZ equivalent)

**Recommended half mask:-**

Particle filtering: EN149:2001 (or AUS/NZ equivalent)

When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

Environmental exposure controls No information available.

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                         |                          |                                          |
|-----------------------------------------|--------------------------|------------------------------------------|
| Appearance                              | Gold                     |                                          |
| Physical State                          | Solid                    |                                          |
| Odor                                    | Odorless                 |                                          |
| Odor Threshold                          | No data available        |                                          |
| pH                                      | No information available |                                          |
| Melting Point/Range                     | 28.5 °C / 83.3 °F        |                                          |
| Softening Point                         | No data available        |                                          |
| Boiling Point/Range                     | 705 °C / 1301 °F         | @ 760 mmHg                               |
| Flash Point                             | No information available | <b>Method -</b> No information available |
| Evaporation Rate                        | Not applicable           | Solid                                    |
| Flammability (solid,gas)                | No information available |                                          |
| Explosion Limits                        | No data available        |                                          |
| Vapor Pressure                          | 0.0075 hPa (145°C)       |                                          |
| Vapor Density                           | Not applicable           | Solid                                    |
| Specific Gravity / Density              | 1.873                    |                                          |
| Bulk Density                            | No data available        |                                          |
| Water Solubility                        | Reacts with water        |                                          |
| Solubility in other solvents            | No information available |                                          |
| Partition Coefficient (n-octanol/water) |                          |                                          |
| Autoignition Temperature                | No data available        |                                          |
| Decomposition Temperature               | No data available        |                                          |
| Viscosity                               | Not applicable           | Solid                                    |
| Explosive Properties                    | No information available |                                          |
| Oxidizing Properties                    | No information available |                                          |
| <b>Other information</b>                |                          |                                          |
| Molecular Formula                       | Cs                       |                                          |
| Molecular Weight                        | 132.91                   |                                          |

## Section 10 - Stability and Reactivity

|                                  |                                                                                                                          |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Reactivity                       | Yes                                                                                                                      |
| Stability                        | Moisture sensitive. Air sensitive.                                                                                       |
| Conditions to Avoid              | Keep away from open flames, hot surfaces and sources of ignition, Incompatible products, Exposure to moist air or water. |
| Incompatible Materials           | Acids, Water, Strong oxidizing agents, Alcohols, Halogens, oxygen, Halogenated compounds, Carbon dioxide (CO2).          |
| Hazardous Decomposition Products | Thermal decomposition can lead to release of irritating gases and vapors. Hydrogen.                                      |
| Hazardous Polymerization         | Hazardous polymerization may occur.                                                                                      |

## Section 11 - Toxicological Information

### Information on Toxicological Effects

|                                                   |                                                                                                                                                                                                                                                            |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Product Information</b>                        | No acute toxicity information is available for this product                                                                                                                                                                                                |
| <b>(a) acute toxicity;</b>                        |                                                                                                                                                                                                                                                            |
| <b>Oral</b>                                       | No data available                                                                                                                                                                                                                                          |
| <b>Dermal</b>                                     | No data available                                                                                                                                                                                                                                          |
| <b>Inhalation</b>                                 | No data available                                                                                                                                                                                                                                          |
| <b>(b) skin corrosion/irritation;</b>             | Category 1 B                                                                                                                                                                                                                                               |
| <b>(c) serious eye damage/irritation;</b>         | Category 1                                                                                                                                                                                                                                                 |
| <b>(d) respiratory or skin sensitization;</b>     |                                                                                                                                                                                                                                                            |
| <b>Respiratory</b>                                | No data available                                                                                                                                                                                                                                          |
| <b>Skin</b>                                       | No data available                                                                                                                                                                                                                                          |
| <b>(e) germ cell mutagenicity;</b>                | No data available                                                                                                                                                                                                                                          |
| <b>(f) carcinogenicity;</b>                       | No data available                                                                                                                                                                                                                                          |
|                                                   | There are no known carcinogenic chemicals in this product                                                                                                                                                                                                  |
| <b>(g) reproductive toxicity;</b>                 | Category 2                                                                                                                                                                                                                                                 |
| <b>(h) STOT-single exposure;</b>                  | No data available                                                                                                                                                                                                                                          |
| <b>(i) STOT-repeated exposure;</b>                | No data available                                                                                                                                                                                                                                          |
| <b>Target Organs</b>                              | No information available.                                                                                                                                                                                                                                  |
| <b>(j) aspiration hazard;</b>                     | Not applicable                                                                                                                                                                                                                                             |
|                                                   | Solid                                                                                                                                                                                                                                                      |
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated. See actual entry in RTECS for complete information                                                                                                                                          |
| <b>Symptoms / effects, both acute and delayed</b> | Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |

## Section 12 - Ecological Information

|                                              |                                                                           |
|----------------------------------------------|---------------------------------------------------------------------------|
| <b>Ecotoxicity effects</b>                   | Reacts with water so no ecotoxicity data for the substance is available.  |
| <b>Persistence and Degradability</b>         |                                                                           |
| <b>Persistence</b>                           | Persistence is unlikely, based on information available.                  |
| <b>Degradability</b>                         | Not relevant for inorganic substances, Reacts with water.                 |
| <b>Degradation in sewage treatment plant</b> | No information available. Water reactive.                                 |
| <b>Bioaccumulative Potential</b>             | Product does not bioaccumulate due to reaction with water                 |
| <b>Mobility</b>                              | Reacts with water. Is not likely mobile in the environment                |
| <b>Endocrine Disruptor Information</b>       | This product does not contain any known or suspected endocrine disruptors |
| <b>Persistent Organic Pollutant</b>          | This product does not contain any known or suspected substance            |
| <b>Ozone Depletion Potential</b>             | This product does not contain any known or suspected substance            |

## Section 13 - Disposal Considerations

**Waste from Residues/Unused Products**

Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**Other Information**

Chemical wastes should be disposed through a licensed commercial waste collection service. Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms.

## Section 14 - Transport Information

**IMDG/IMO**

UN-No UN1407  
Proper Shipping Name CAESIUM  
Hazard Class 4.3  
Packing Group I

**ADG**

UN-No UN1407  
Proper Shipping Name CAESIUM  
Hazard Class 4.3  
Packing Group I

| Component                   | Hazchem Code |
|-----------------------------|--------------|
| Cesium<br>7440-46-2 ( >95 ) | 4W           |

**IATA**

UN-No UN1407  
Proper Shipping Name CAESIUM  
Hazard Class 4.3  
Packing Group I

Environmental hazards No hazards identified

Special Precautions No special precautions required

Additional information None known

## Section 15 - Regulatory Information

**Safety, health and environmental regulations/legislation specific for the substance or mixture****National Regulations** Australia

See section 8 for national exposure control parameters.

**Standard for the Uniform Scheduling of Medicines and Poisons**

No poison schedule number allocated.

#### Australian Industrial Chemicals Introduction Scheme (AICIS)

| Component          | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information |
|--------------------|-------------------------------------------------------------|------------------------|
| Cesium - 7440-46-2 | Present                                                     | -                      |

#### Australian - Illicit Drug Precursors/Reagents Substance List

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

#### Chemicals of Security Concern

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

**National pollutant inventory** Not applicable

#### Prohibition or notification/licensing requirements

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

#### International Inventories

| Component | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL     |
|-----------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|----------|
| Cesium    | X    | -     | 231-155-4 | -      | X    | -   | X    | X     | X    |      | X     | KE-05428 |

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

#### International Regulations

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

#### Basel convention on the control of transboundary movements of hazardous wastes and their disposal

Not applicable.

| Component | CAS No    | OECD HPV       | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-----------|-----------|----------------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Cesium    | 7440-46-2 | Not applicable | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |

**Authorisation/Restrictions according to EU REACH** Not applicable



## Section 16 - Other Information

### Legend

|                                                                                                      |                                                                                                                              |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>AICS</b> - Australian Inventory of Chemical Substances                                            | <b>NZIoC</b> - New Zealand Inventory of Chemicals                                                                            |
| <b>TSCA</b> - United States Toxic Substances Control Act Section 8(b) Inventory                      | <b>EINECS/ELINCS</b> - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances |
| <b>DSL/NDL</b> - Canadian Domestic Substances List/Non-Domestic Substances List                      | <b>ENCS</b> - Japanese Existing and New Chemical Substances                                                                  |
| <b>IECSC</b> - Chinese Inventory of Existing Chemical Substances                                     | <b>KECL</b> - Korean Existing and Evaluated Chemical Substances                                                              |
| <b>PICCS</b> - Philippines Inventory of Chemicals and Chemical Substances                            | <b>CAS</b> - Chemical Abstracts Service                                                                                      |
| <b>TWA</b> - Time Weighted Average                                                                   | <b>ACGIH</b> - American Conference of Governmental Industrial Hygienists                                                     |
| <b>IARC</b> - International Agency for Research on Cancer                                            | Predicted No Effect Concentration (PNEC)                                                                                     |
| <b>ICAO/IATA</b> - International Civil Aviation Organization/International Air Transport Association | <b>IMO/IMDG</b> - International Maritime Organization/International Maritime Dangerous Goods Code                            |
| <b>MARPOL</b> - International Convention for the Prevention of Pollution from Ships                  | <b>ADG</b> - Australian Code for the Transport of Dangerous Goods by Road and Rail                                           |
| <b>NZS 5433:2020</b> - Transport of Dangerous Goods on Land                                          | <b>OECD</b> - Organisation for Economic Co-operation and Development                                                         |
| <b>LD50</b> - Lethal Dose 50%                                                                        | <b>LC50</b> - Lethal Concentration 50%                                                                                       |
| <b>EC50</b> - Effective Concentration 50%                                                            | <b>ATE</b> - Acute Toxicity Estimate                                                                                         |
| <b>WEL</b> - Workplace Exposure Limit                                                                | <b>RPE</b> - Respiratory Protective Equipment                                                                                |
| <b>DNEL</b> - Derived No Effect Level                                                                | <b>NOEC</b> - No Observed Effect Concentration                                                                               |
| <b>POW</b> - Partition coefficient Octanol:Water                                                     | <b>BCF</b> - Bioconcentration factor                                                                                         |
| <b>vPvB</b> - very Persistent, very Bioaccumulative                                                  | <b>PBT</b> - Persistent, Bioaccumulative, Toxic                                                                              |
| <b>VOC</b> - (Volatile Organic Compound)                                                             |                                                                                                                              |

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

### Revision Date

22-Nov-2024

### Revision Summary

Update to GHS format.

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

## End of Safety Data Sheet